

Edoardo Mosca, PhD

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 [Edoardo Mosca](#) |  [EdoardoMosca](#) |  [Edoardo Mosca](#)

Technical Lead for ML Research and Engineering that can speak the product language. Looking for roles needing strong drivers and owners. Background in maths and PhD in CS (~20 publications). Several years of experience between academia, corporate, and startups.

EXPERIENCE

Apr 2024 - Current

ML Research Scientist

Liquid AI (MIT), Boston, United States (Remote)

- Liquid foundation models (LFMs) core research/engineering team, developed core modules for general sequence modeling and their integration in downstream internal/external packages.
- Model customization lead: post-training, alignment, and synthetic data generation.
- Main responsible for LLMs and VLMs on-prem customization tools. Model integration, data validation and loading, distributed training and inference, packaging and obfuscation.
- Brought code to production and shipped, led several large customer engagements.

Jan 2022 - Dec 2023

NLP Project Lead

Software Campus Program, Munich, Germany

- Awarded 100k from the German Ministry to lead a team of ten people.
- Developed IFAN, a platform to interactively steer LLMs online at various stages of deployment and align them with human intent.
- Used adapters to efficiently store feedback representations for the internal model library
- In collaboration with the Holtzbrinck Publishing Group.

Sep 2022 - Aug 2023

Co-Founder and CTO

Cryptoscape.ai, Berlin, Germany

- Technical lead in building an LLM-based automated investment advisor for ETFs and crypto. Team of four.
- All things backend: scraping, database, query understanding, RAG.
- Our work has been supported and funded by KIEZ Berlin.

Nov 2018 - Apr 2020

ML Research And Development Assistant

Infineon Technologies, Munich, Germany

- Application of computer vision and non-standard representation learning to estimate and optimize SoC hardware design metrics.
- DenseNets and ResNets ablations for estimation. Non-differential optimization and RL for optimization.
- Results led to three scientific publications, which where I co-authored as the 1st or 2nd author.

Feb 2018 - Oct 2018

Data Scientist and Software Developer

ParkHere, Munich, Germany

- Design, development, and testing of data-driven algorithms for the automatic detection of pedestrians and cars in large-scale commercial parking lots. Heuristics + classic ML solutions (clustering + SVM).
- Productionized 1st version in python, 2nd version in C++

Stack and Expertise

Languages: Python  (+ Pandas, Numpy, Scikit-learn, etc..) | can refresh   

ML/NLP: Pytorch  | Hugging Face  | LangChain  | LLama Index  | Tensorflow-Keras 

Backend/BaaS: FastAPI  | Supabase  **OS / Version Control:** Linux (bash)  | Git 

Databases: PostgreSQL  (+ pgvector) | MongoDB 

EDUCATION

Aug 2020 – Sep 2023

Doctor in NLP and Explainable AI, Computer Science

Technical University of Munich, Germany

- Explored the usage of post-hoc explainability to work towards more interpretable, robust, and (human) controllable NLP models. ~ 12 papers including ACL, COOLING, NAACL, ECML. • **Supervisor:** Prof. Dr. Georg Groh.
- **Dissertation:** *Explainable AI for the Human-Centric Development of NLP Models*. • **Grade:** Highest Honors

Oct 2017 – Apr 2020

M.Sc. Mathematics in Data Science

Technical University of Munich, Germany

- Focus on machine learning from both the mathematical (theoretical) and computer science (hands-on) sides. • Practical experiences in vision and NLP. • **Thesis:** *Explainability of Hate Speech Detection Models*.

Sep 2013 – May 2017

B.Sc. Mathematics

University of Pisa, Italy

- Minor in computer science. • **Thesis:** *Computational Class. for Regular Expression Membership Testing*

SELECTED PUBLICATIONS

ACL
2022

Edoardo Mosca et al. *“That Is a Suspicious Reaction!”: Interpreting Logits Variation to Detect NLP Adversarial Attacks.*

COLING
2022

Edoardo Mosca et al. *SHAP-Based Explanation Methods: A Review for NLP Interpretability.*

AAACL
2023

Edoardo Mosca et al. *IFAN: An Explainability-Focused Interaction Framework for Humans and NLP Models*

MDPI
2023

Edoardo Mosca et al. *Distinguishing Fact from Fiction: A Benchmark Dataset for Identifying Machine-Generated Scientific Papers in the LLM Era. (TrustNLP Extension)*

CERTIFIED TRAININGS

○ Agile for Management

IAV, Berlin, 2 days, Aug 2022

○ Customer-Centric Design Thinking

Grundfos, Denmark, 2 days, Sep 2022

Workplace Personalities

○ Merck, Darmstadt, 2 days, Nov 2022

○ Hiring and Talent Acquisition

Zeiss, Oberkochen, 2 days, Nov 2022

○ Change Management

Trumpf, Stuttgart, 2 days, May 2023

Culture & Sustainability

○ Huawei, Munich, 1 day, June 2023

LANGUAGES

Italian (Native) | English (C2) | German (C1) | Spanish (C1) | French (A2)

REFERENCES

Prof. Dr. Georg Groh (TUM)
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Dr. Lorenzo Servadei (Sony AI)
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Dr. Robert Seidl (Stats Perform)
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